

INTRODUCTION

MeterIQ

Explainable Smart Meter Analytics Platform

An Integrated Anomaly Detection & Load-Forecasting Platform for Indian Electricity Distribution Networks

DATASET

Real Indian Smart Meter Data
(Mathura & Bareilly)

CORE MODULES

1. Anomaly Detection
2. Load Forecasting

Introduction → Motivation → Problem Statement → Solution

The project uses smart-meter data to identify unusual consumption and forecast future load.

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MOTIVATION

Why Is This Project Needed?



High Non-Technical Losses

Electricity theft, meter tampering and abnormal consumption can cause losses and require manual investigation.



Unexpected Demand Peaks

Unexpected demand peaks can put extra load on the distribution system. Forecasting can help utilities prepare for these changes.



Underused Smart-Meter Data

Smart meters generate large amounts of time-based electricity data that can be used for better analysis and planning.

Smart meters already provide detailed electricity data. The challenge is turning this data into useful information for forecasting and investigation.

Key Challenges in Smart Meter Analytics



Anomaly Detection

- ▶ Smart meters can show unusual consumption patterns.
- ▶ It is difficult to know whether an unusual reading is normal or needs investigation.
- ▶ Alerts should provide useful information instead of only marking a reading as abnormal.



Load Forecasting

- ▶ Smart-meter readings need to be cleaned and processed before forecasting.
- ▶ Electricity demand changes with time, day and other conditions.
- ▶ Forecasting should be connected with anomaly detection instead of working separately.

Root cause: Smart-meter data is available, but it is difficult to turn that data into reliable forecasts and useful anomaly alerts in one system.

OUR SOLUTION

One Platform, Two Honest, Explainable Modules



Anomaly / Theft-Risk Detection

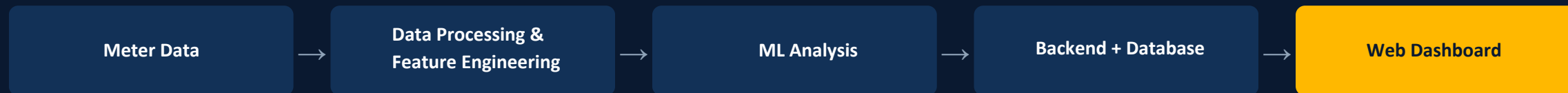
- Analyse meter behaviour and identify unusual patterns
- Detect abnormal consumption and meter behaviour that may indicate tampering or theft
- Output: Normal → Suspicious → High-Risk with explainable alerts



Short-Term Load Forecasting

- Use historical meter data to predict future demand
- Include time-based and weather-related information
- Compare expected and actual load to identify deviations

End-to-End Flow



The system will combine meter analytics, forecasting and anomaly investigation through a web-based dashboard for utility decision-making.

Tech Stack: React.js • Flask • Python • PostgreSQL • Pandas • Scikit-learn • Chart.js • Git/GitHub